ELEVATE LABS TASK 5
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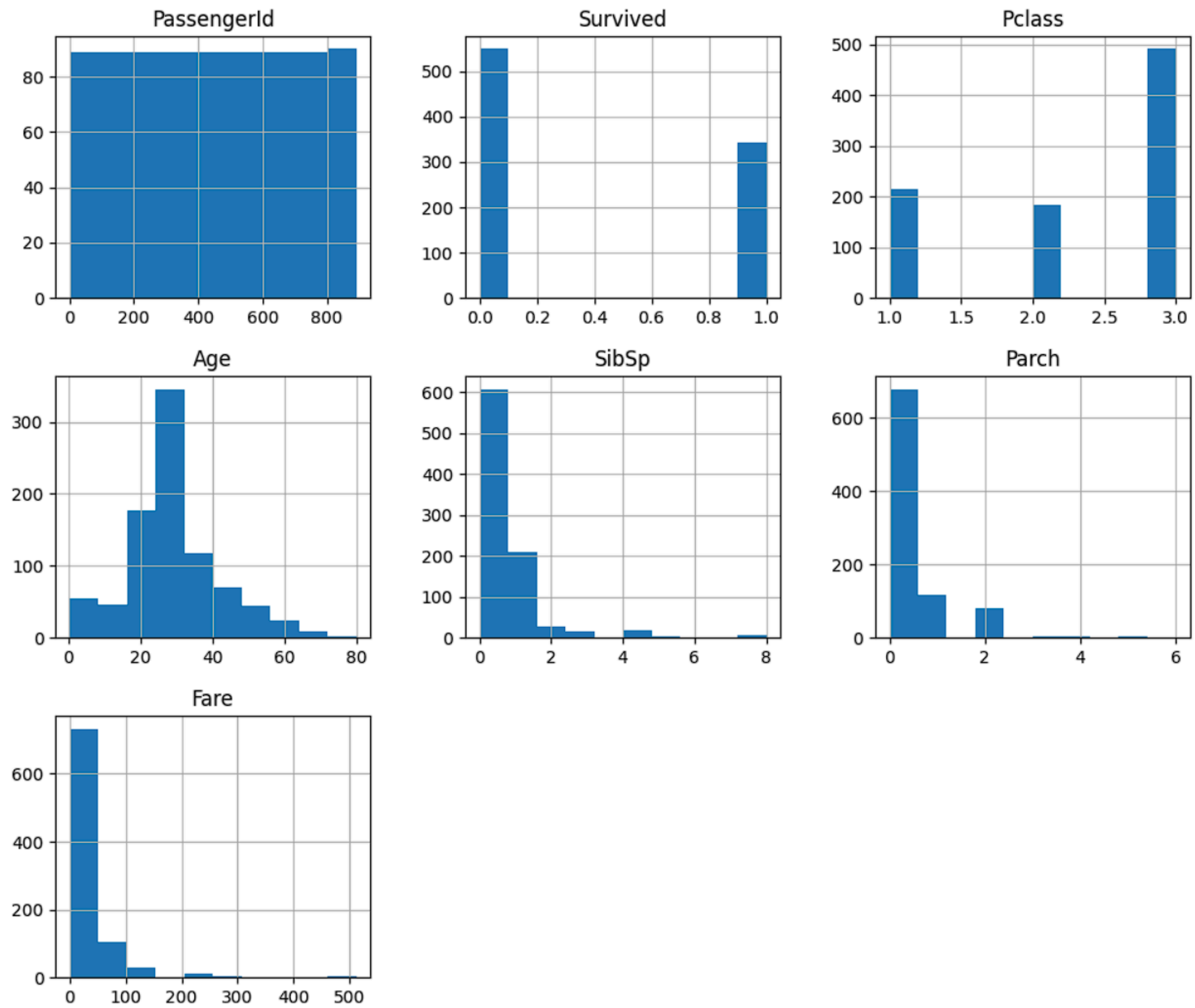
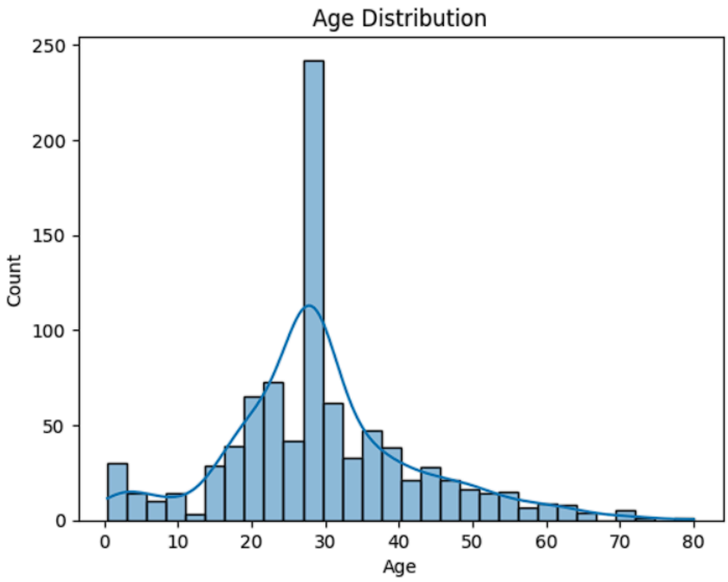
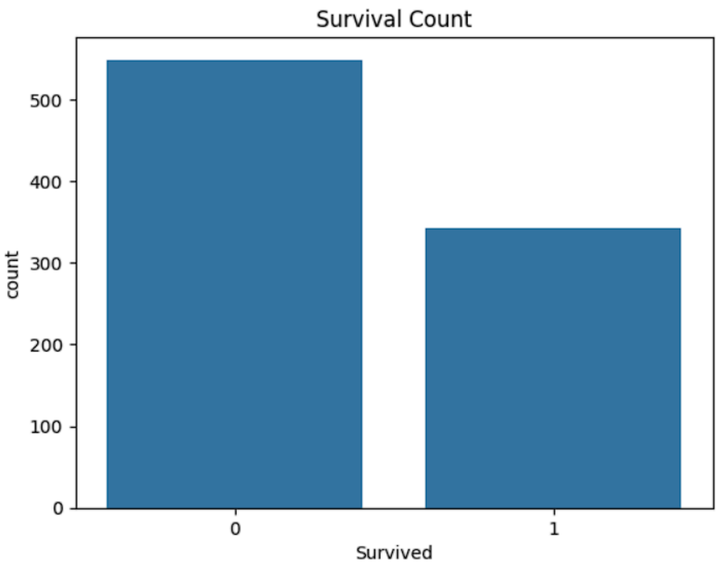
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=pd.read_csv('train.csv')
df.value_counts()
df.info()
df.describe()
df.isnull().sum()

df['Age'].fillna(df['Age'].median(), inplace=True)
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
df.drop(columns=['Cabin'], inplace=True)
df.head()

df.hist(figsize=(12,10))
plt.show()

sns.histplot(df['Age'], kde=True)
plt.title("Age Distribution")
plt.show()

sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()

sns.countplot(x='Sex', hue='Survived', data=df)
plt.title("Survival by Gender")
plt.show()

sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title("Survival by Passenger Class")
plt.show()

sns.boxplot(x='Survived', y='Age', data=df)
plt.title("Age vs Survival")
plt.show()
```

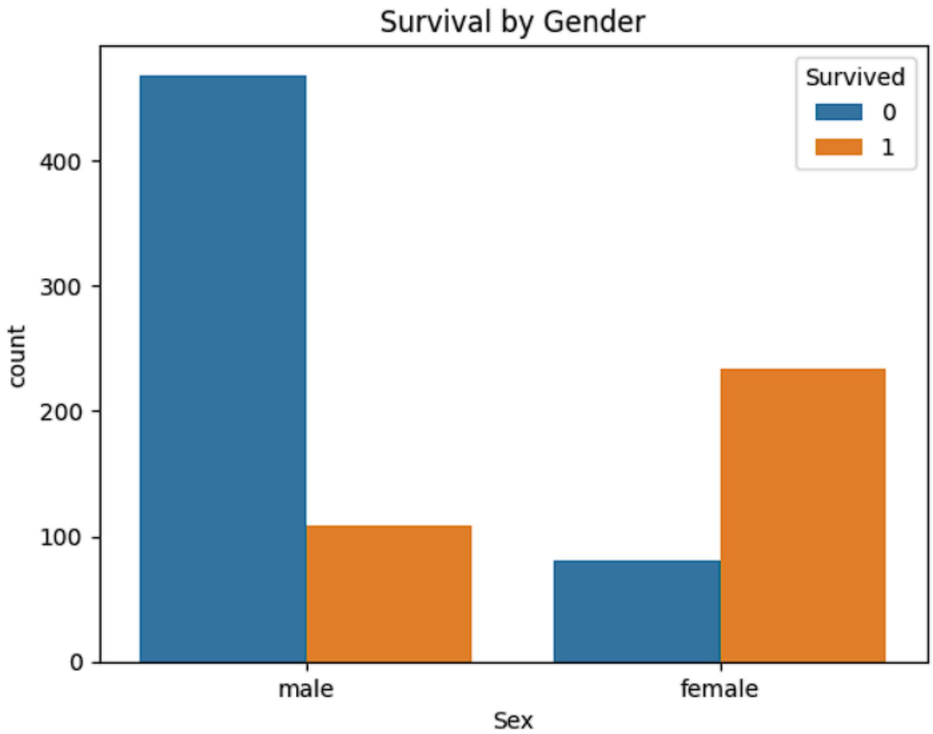
```python
sns.scatterplot(x='Fare', y='Pclass', hue='Survived', data=df)
plt.title("Fare vs Passenger Class")
plt.show()

pair_df = df[['Survived','Age','Fare','Pclass']]
sns.pairplot(pair_df, hue='Survived')
plt.show()

temp = df.copy()
temp['Sex'] = temp['Sex'].map({'male':0,'female':1})
temp['Embarked'] = temp['Embarked'].map({'S':0,'C':1,'Q':2})
corr = temp.corr(numeric_only=True)
plt.figure(figsize=(10,8))
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()

print("\nSUMMARY")
print("\ntotal passengers:", len(df))
survival_rate = df['Survived'].mean()*100
print("\nSurvival Rate:", round(survival_rate,2),"%")
print("\nAverage Age:",df['Age'].mean())
print("\nAverage Fare:",df['Fare'].mean())
```

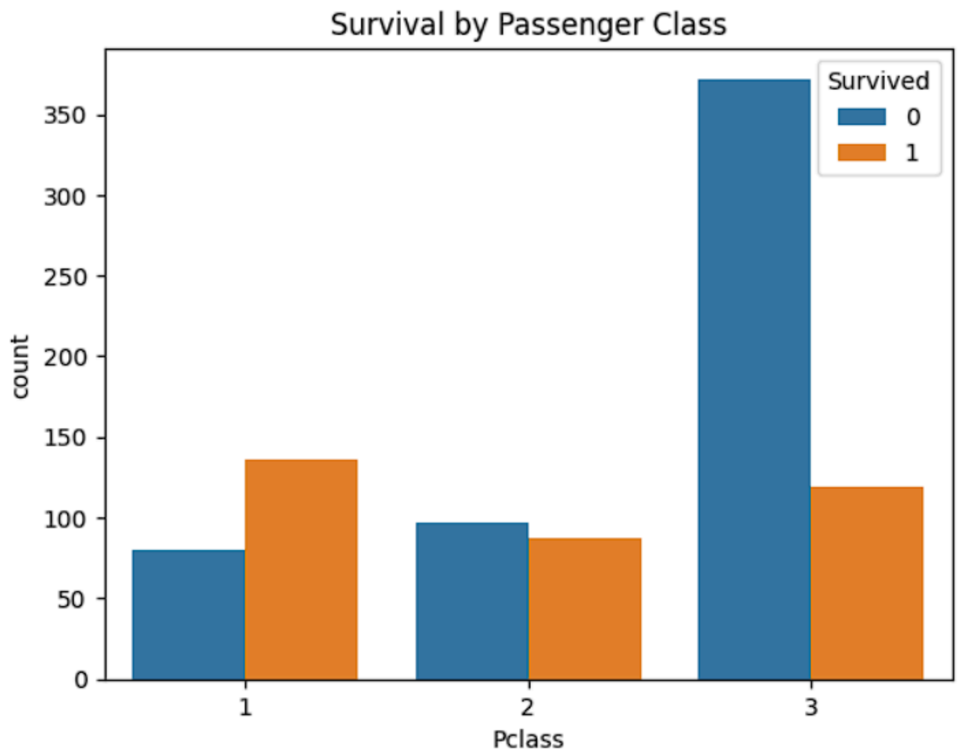
```
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
```

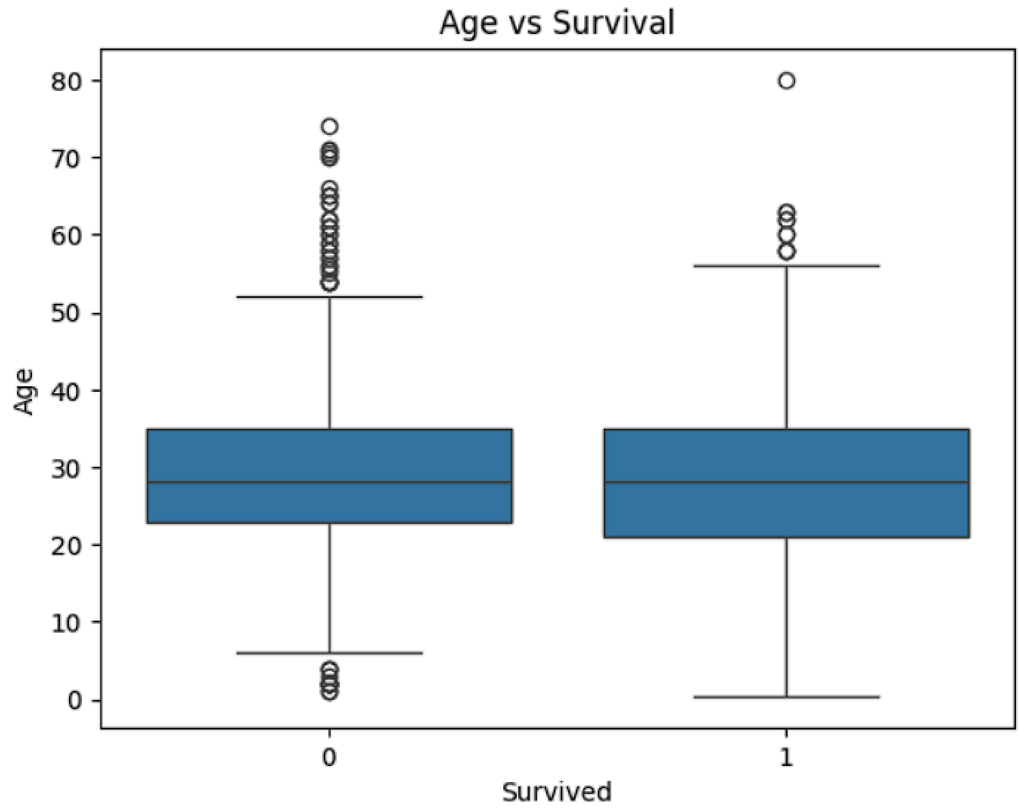


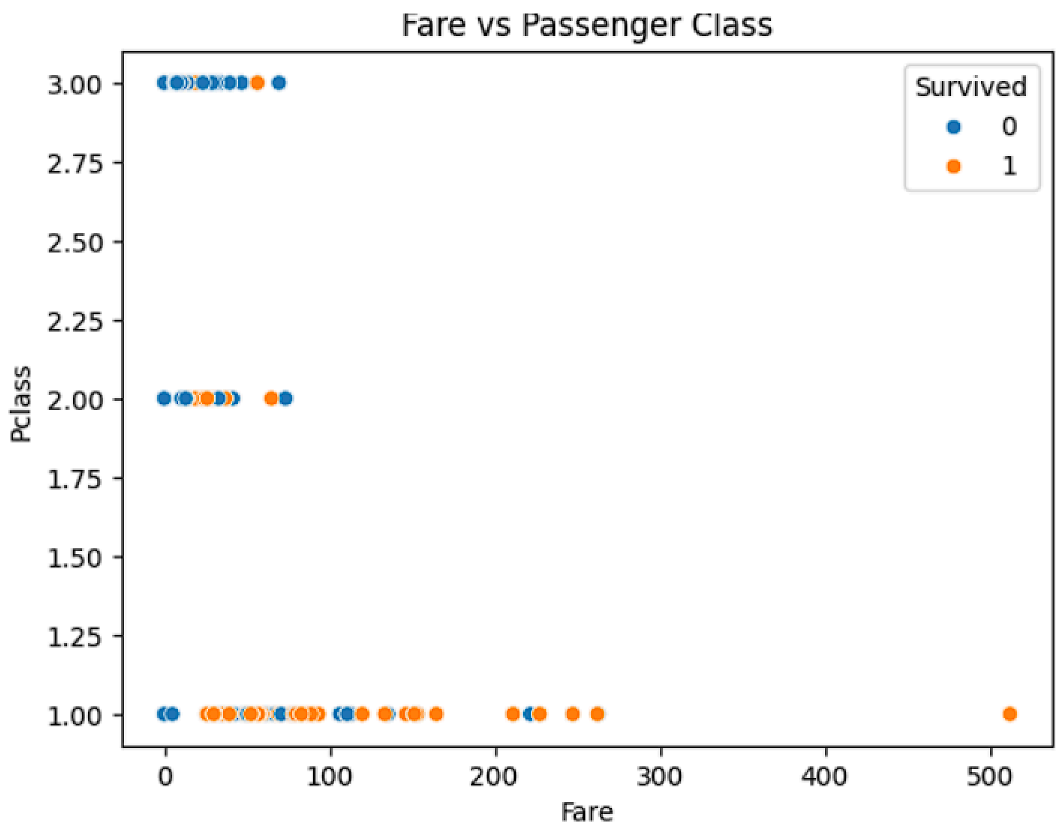

Age Distribution
Count
Age
250
200
150
100
50
0
0
10
20
30
40
50
60
70
80

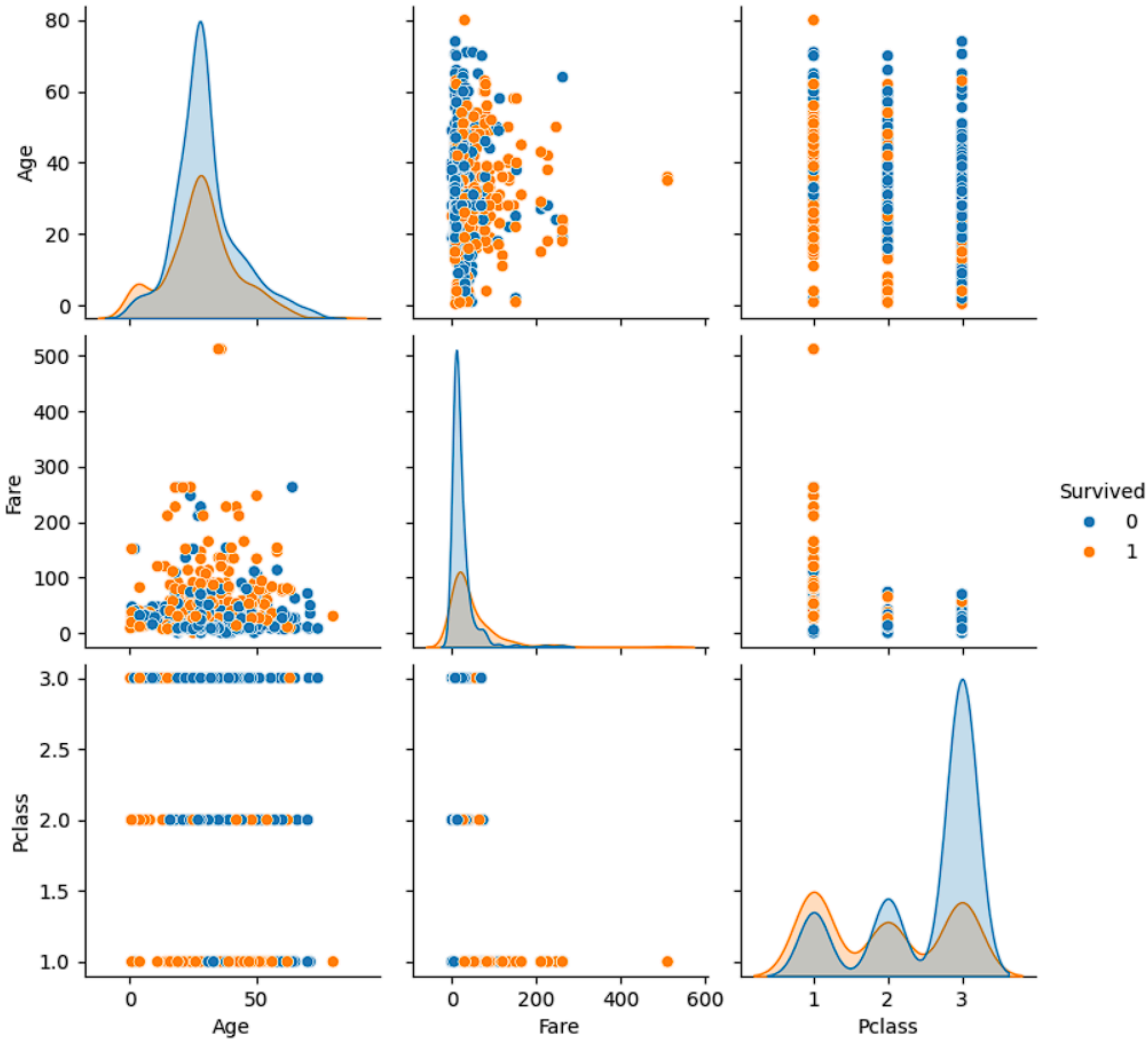

Survival Count
count
Survived
500
400
300
200
100
0
0
1


Survival by Gender
Survived
0
1
count
400
300
200
100
0
male
female
Sex

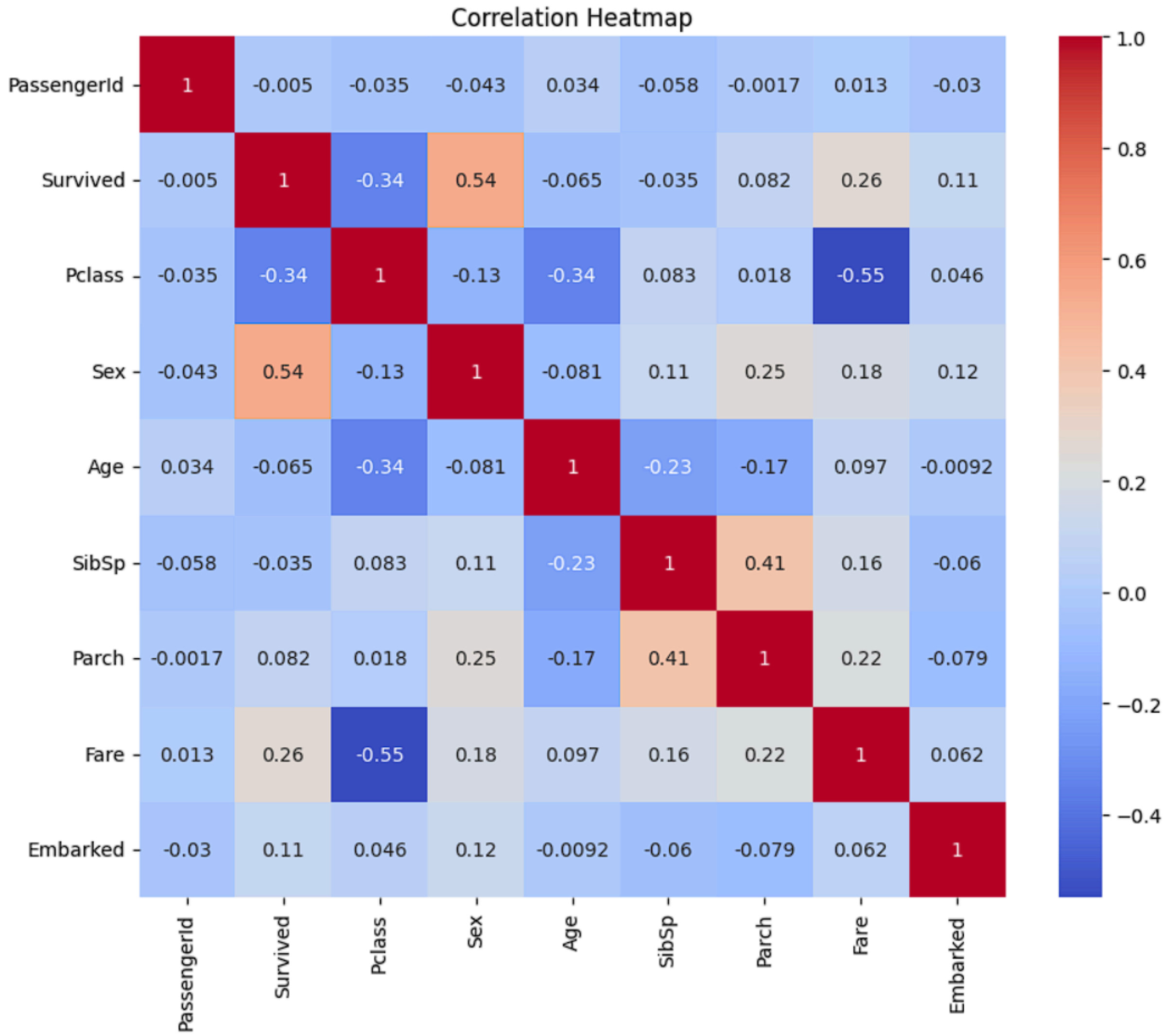

Survival by Passenger Class
Survived
0
1
count
350
300
250
200
150
100
50
0
1
2
3
Pclass


Age vs Survival
80
70
60
50
40
30
20
10
0
Age
0
1
Survived


Fare vs Passenger Class
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00
Pclass
Survived
0
1
0
100
200
300
400
500
Fare



Age
Fare
Pclass
Survived
0
1




```
SUMMARY

total passengers: 891

Survival Rate: 38.38 %

Average Age: 29.36158249158249

Average Fare: 32.204207968574636
```